Finite-Prefix Terras Parity-Vector Stopping-Time Density Bounds in Lean 4

Benjamin Stanley Frohman
Independent Researcher | ORCID: 0009-0006-7068-3718
frohmanbenjamin@gmail.com

Origin Provenance: October 4, 2026 (15:45:31 CDT)
Repository: BenFrohman/collatz-terras-density | Licensed under Apache-2.0

Abstract

We present a zero-sorry, 7-module Lean 4 formalization (TerrasDensity/Basic.lean, Covering.lean, Density.lean, Inclusion.lean, Tail.lean) of the finite-prefix Terras parity-vector stopping-time density structure for the $3n+1$ map. Here FiniteStoppingTime n denotes the existence of $k \geq 1$ such that $\mathrm{iter}(k, n) < n$. The module Tail.lean formalizes the binomial tail count of bad parity words of length k, and Density.lean links residue class inclusion to the asymptotic density-one stopping-time statement, while strictly separating Terras density-one finite stopping time from the open global Collatz convergence claim ($\forall n > 0$, $\exists k$, $T^k(n) = 1$).

1 Architecture and Provenance

- **Origin Commit Timestamp:** Sunday, October 4, 2026, 15:45:31 CDT (2026-10-04T20:45:31Z).
- **Modules:** TerrasDensity.lean, TerrasDensity/Basic.lean, TerrasDensity/Covering.lean, TerrasDensity/Density.lean, TerrasDensity/Inclusion.lean, TerrasDensity/Tail.lean.
- **Honest Scope:** Proves the structural Terras parity-vector inclusion and binomial tail bounds with zero sorry. Does not claim that density-one finite stopping time implies that every positive integer reaches 1.

License and Copyright

Copyright © 2026 Benjamin Stanley Frohman. All rights reserved. Licensed under the Apache License, Version 2.0 (http://www.apache.org/licenses/LICENSE-2.0).

A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/collatz-terras-density
- Verified commit:
  5716a1c12568299742be9ea35f9a0b40d141fa21

- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/collatz-terras-density/actions/runs/41570505495`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/collatz-terras-density/archive/`
  `5716a1c12568299742be9ea35f9a0b40d141fa21.tar.gz`
  `b86807210a58f26059b6380451d65a7f0a3a067424cd078d9910269434561f9f`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 9 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`TerrasDensity.not_neverDrops_iff`	`[propext, Classical.choice, Quot.sound]`
`TerrasDensity.covering_from_assumption`	`(no axioms)`
`TerrasDensity.density_from_assumption`	`[propext, Quot.sound]`
`TerrasDensity.density_of_covering`	`[propext, Quot.sound]`
`TerrasDensity.drop_of_affine`	`[propext, Classical.choice, Quot.sound]`
`TerrasDensity.inclusion_of_affine`	`[propext, Classical.choice, Quot.sound]`
`TerrasDensity.terras_inclusion`	`[propext, Classical.choice, Quot.sound]`
`TerrasDensity.tail_le`	`[propext, Quot.sound]`
`TerrasDensity.two_pow_le_three_pow`	`(no axioms)`